Lesson plan · for the teacher

Big problems, good answers

Optimization · Level 4 · Lesson 4

Time
40 min

Materials
Board and chalk; A name card for each pupil in one group (about 12); A seating grid drawn on the board

Learning goals

- Improve a plan in small steps, and see why it can get stuck.
- Escape a stuck plan by allowing some worse moves.
- Decide when good enough is enough, and plan a project with people.

Activity: Swap to improve

1. Warm-up (5 min): Ask pupils to guess how many ways 12 pupils can sit at 12 desks. Reveal 479,001,600, and explain why nobody checks them all.
2. Set up (7 min): Draw 12 desks in 3 rows of 4. Agree pairs who work together (each pair sitting apart costs the number of steps between their desks) and chatty pairs (side by side costs 4).
3. Climb (10 min): Groups place name cards at random, work out the cost, then swap two cards at a time, keeping a swap only if the cost goes down. Stop when no swap helps and write the cost.
4. Escape (10 min): Now groups may keep up to three swaps that make the cost a little worse, early on only, then go back to better-only swaps. Write the new cost.
5. Wrap-up (8 min): Compare costs across groups. Who got stuck? Did the worse swaps help? Agree what cost would be good enough and why.

Prepare the pairs in advance on a card (who works together, who is chatty) so groups score plans the same way. Use made-up names if pupils might be upset by who sits where.

Discussion

- Why can a plan be stuck even though no swap improves it?
- Who should decide when our timetable is good enough?
- What rules might our data not know about?

Exit ticket

- Write the first two steps you would take to lead an optimization project at our school.

Try it at school tomorrow
Draw a seating plan on the board with pupils' names on cards, and agree a simple cost (for example, one point for every pair of friends who work together sitting apart). Pupils swap two cards at a time and keep a swap only if the cost goes down. When they're stuck, allow one worse swap and see whether they find a better plan.

Practice questions · for you

Big problems, good answers

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1 **The solver for next week's exam invigilation rota has run for an hour. Its best rota is within 2% of a bound no rota can beat, and the rota is due tomorrow morning. Do you keep it running?**

◯ Yes: run until it proves the best ◯ Start again from scratch

◯ No: 2% is good enough here; check it with people

2 **Put the steps of a school optimization project in order.**

☐ Agree the goal and the fairness rules, before any plan is seen

☐ Agree the problem with the people it affects

☐ Pilot it, explain it and decide together

☐ Collect the data and build a small model

☐ Test the model on last year's plan

3 **Complete the simulated annealing rule: always keep a swap that isn't worse, sometimes keep a worse one, and cool down slowly.**

```
d = new_cost - cost
r = random.random()
if d <= __________ or r < math.__________(-d / T):
    plan, cost = new_plan, new_cost
T = T * __________
```

Word bank: `exp` `0` `log` `1.5` `1` `0.999`

4 **If hill climbing stops and no single swap improves the plan, the plan is the best possible.**

◯ True ◯ False

5 **You are leading an optimization of next term's duty rota. Which of these should you decide before you start the solver? Choose all that apply.**

☐ To hide the rules so nobody games them ☐ The goal and the fairness rules, agreed with staff

☐ How long the solver may run, and what gap is good enough

☐ Who makes the final call, and how to appeal ☐ Which teacher should get the worst duty

✎ **Which plan at your school is too big to check by hand: the timetable, invigilation, the bus? What would good enough mean for it, and who should decide that?**

Check your answers

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1 **No: 2% is good enough here; check it with people**

The bound says no rota can be more than 2% better than the one already found.

2 **Top to bottom: 2, 1, 5, 3, 4**

People and the problem come first, and the goal and fairness rules are agreed before anyone can see who wins.

3

```
d = new_cost - cost
r = random.random()
if d <= 0 or r < math.exp(-d / T):
    plan, cost = new_plan, new_cost
T = T * 0.999
```

d <= 0 keeps every swap that doesn't raise the cost.

4 **False**

It is only the best nearby: a local best.

5 **The goal and the fairness rules, agreed with staff; How long the solver may run, and what gap is good enough; Who makes the final call, and how to appeal**

Goals, fairness rules, stopping rules and who decides are all agreed in advance, so the result can't be bent afterwards.